Pothuri Devi Amrutha

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	7.66
Aditya Junior College ,Palakollu	Class XII (Board of Intermediate Education Andhra Pradesh)	2024	97.30%
Aditya Public School, Palakollu	Class X (Board of Secondary Education,Andhra Pradesh)	2022	90.83%

PROJECTS

Arduino-Based Diode I-V Characterization System *Arduino* | *Embedded C/C++* | *Python* | *Data Acquisition*

- Designed and developed an Arduino-based automated I-V characterization system for semiconductor devices, enabling controlled voltage sweeping and real-time measurement of device voltage and current.
- Characterized the electrical behavior of 1N4003, Zener, Schottky diodes, and solar cells under forward- and reverse-bias conditions to obtain their I-V characteristics.
- Developed embedded firmware for voltage control, sensor interfacing, data acquisition, and serial communication between the measurement hardware and computer.
- Built a Python-based visualization and analysis tool for real-time plotting of I-V curves and comparison of electrical characteristics across different semiconductor devices.

Single-Axis Solar Tracking System *Embedded Systems* | *Sensor Interfacing* | *Servo Control* | *Control Systems*

- Designed and implemented a single-axis solar tracking system using LDR sensors, ESP32/Arduino, and a servo motor to automatically orient the solar panel toward the direction of maximum incident light.
- Developed a feedback-based control algorithm that continuously compares LDR sensor readings and adjusts the panel position to track changes in the sun's position.
- Integrated LDR sensors with the microcontroller for real-time light-intensity measurement and implemented servo motor control for accurate angular positioning.
- Evaluated the tracking system against a fixed-orientation setup to demonstrate improved alignment with incident sunlight and enhanced solar energy capture.

Lock-In Amplifier for Weak Signal Detection *Analog Circuit Design* | *LTspice* | *Altium Designer* | *PCB Design*

- Designed and simulated an analog lock-in amplifier for detecting low-amplitude signals buried in noise using phase-sensitive detection and synchronous demodulation.
- Implemented signal modulation, analog multiplication using AD633, amplification using ADA4610, and Sallen-Key low-pass filtering for effective signal extraction and improved SNR.
- Designed and fabricated the complete PCB in Altium Designer, including schematic capture, component placement, routing, grounding, and power distribution for low-noise operation.
- Validated the fabricated prototype through hardware testing and compared experimental measurements with LTspice simulation results to evaluate circuit performance.

Traffic Signal Controller for a T-Junction *Verilog HDL* | *RTL Design* | *Moore FSM* | *Functional Verification*

- Designed an RTL-based traffic signal controller for a T-junction using a Moore finite state machine (FSM), defining dedicated states for vehicle flow, transition intervals, pedestrian crossing, and emergency handling.
- Implemented synchronous timer-based sequencing using counters to control signal durations and ensure safe transitions between traffic states.
- Integrated emergency vehicle priority with state backup and recovery, along with pedestrian crossing control and buzzer alerts for coordinated traffic management.
- Developed synthesizable Verilog RTL and verified the controller using functional testbenches and waveform analysis, covering normal traffic operation, pedestrian requests, and emergency conditions.

POSITIONS OF RESPONSIBILITY

Core Member (2025 - 2026) of ***Elektronica Club*** (Electronics and Signal Processing club), IIT Hyderabad

SKILLS

Programming : C, Python, Verilog,System Verilog, MATLAB

EDA & Design Tools: LTSpice, Altium Designer, KiCad, SolidEdge, cadence,LaTeX

Areas: Analog Electronics, Digital Design, Signal Processing

RELEVANT COURSES

Relevant Courses : Digital Systems, Signals & Systems, Control Theory for Circuits, Circuits and Networks, semiconductor Devices,Analog Electronics, Computer Architecture,Intro to CMOS processing.

Maths and software courses : Data Structures and Algorithms,Probability and Random Processes,Machine Learning,Intro to statistics,complex variables.

Labs : Electric Circuits Lab, Semiconductor Devices Lab, PCB Design, Digital Fabrication, Electronic System Design Lab

EXTRACURRICULAR

NSS Volunteer -- engaged in community service and social initiatives

Pursuing painting as a creative hobby, exploring different mediums, styles, and techniques.